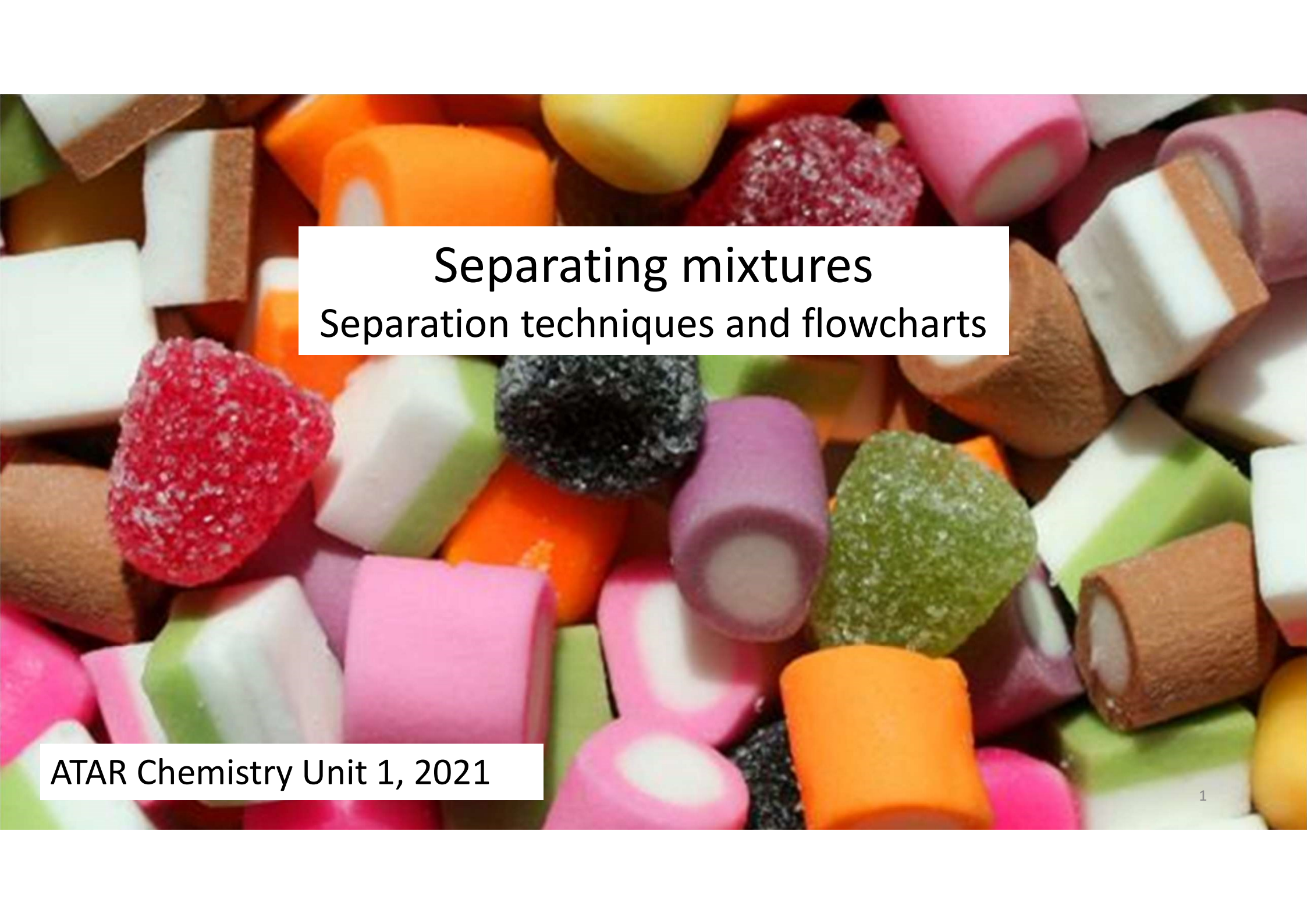




Separating mixtures

Separation techniques and flowcharts

ATAR Chemistry Unit 1, 2021



Outline

Outline

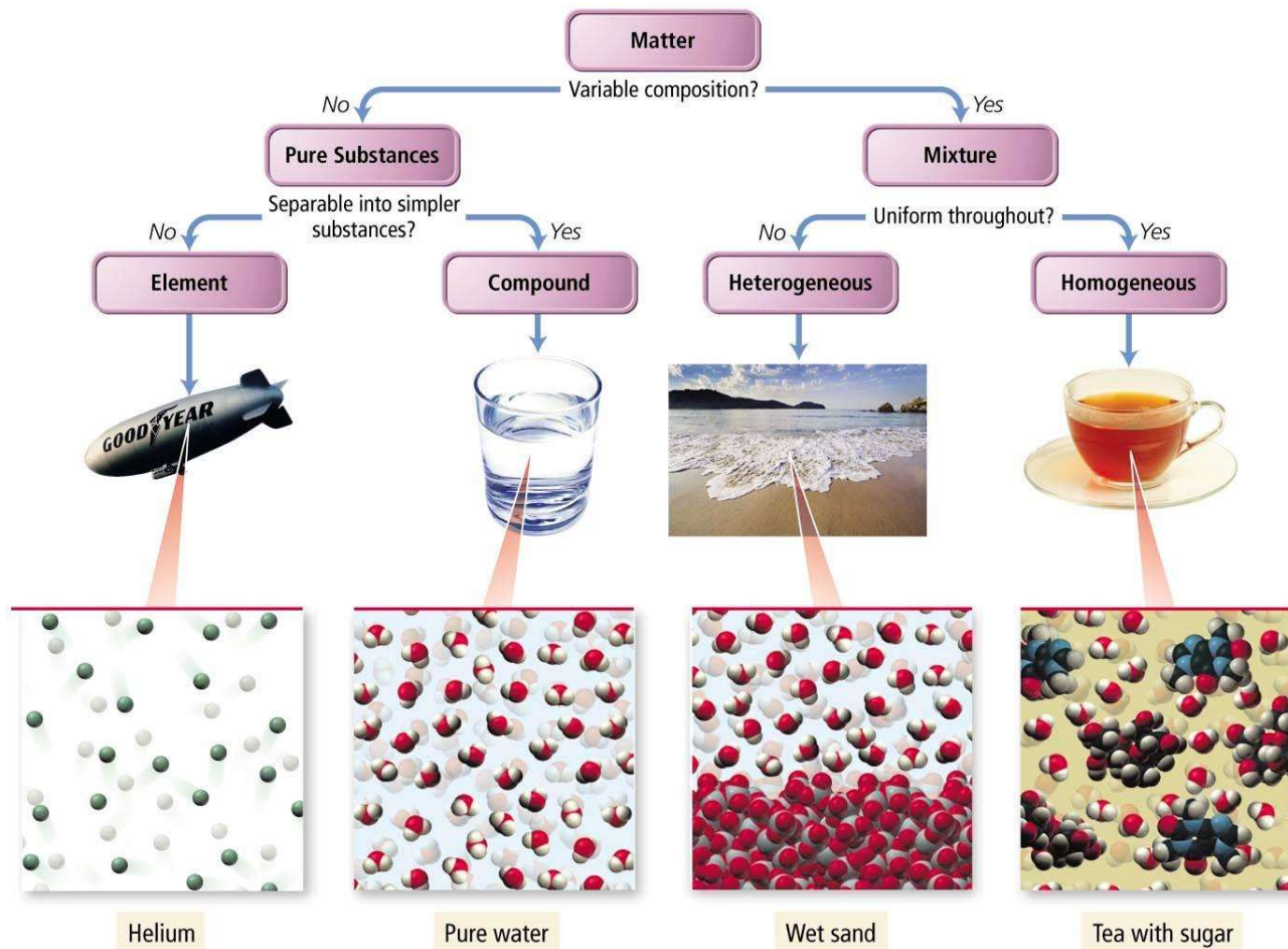
- Physical properties
- Separation techniques

Resources:

- Pearson ch 1.3 review Q 1.3
- STAWA



Review – classification of matter





Review – physical properties

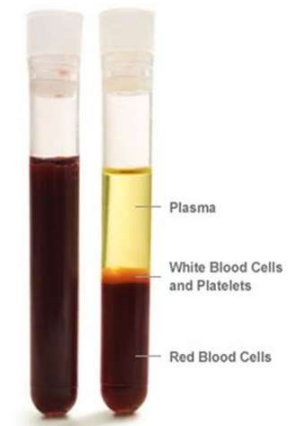
- Melting point – the temperature at which the solid changes state to a liquid
- ★ • Boiling point – the temperature at which the vapour pressure of the liquid equals the surrounding external pressure. The liquid changes into a vapour.
- Reactivity – the tendency of a substance to undergo chemical reactions, either with itself or with other materials.
- Hardness – the ability of a material to resist deformation from a force
- ★ • Density – a measure of a materials mass per volume
- ★ • Solubility – the ability for a one substance to dissolve in another substance
- ★ • Magnetism – the ability attract/repel other magnetic materials



Purifying materials

Separation and purification techniques play an important role in industry, research and at home, for example:

- Crude oil – can be separated into a large number of components using fractional distillation, the components being used in cars, to make plastics etc.
- Purifying silicon – 99.999 % silicon is used in solar panels, but you need 99.9999999 % silicon to make a microchip!
- In the kitchen – making cheese requires us to separate curds (solid, fatty bit) from the whey (the water like bit)
- Medicine – separate blood into plasma and blood cells





Purifying materials

What physical property can we use to separate this mixture?



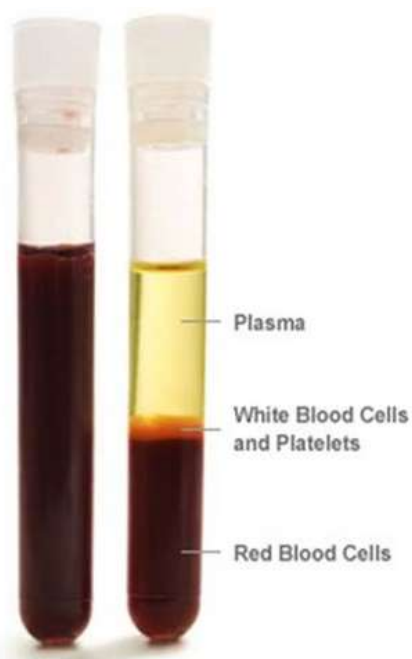
Crude oil

- Mixture of different hydrocarbons and organic compounds
- Can be separated based on the different boiling points of the substances
- Technique used: fractional distillation (we will cover in detail later)



Purifying materials

What physical property can we use to separate this mixture?



Blood

- Observation: See in the pictures that the darker red substance sinks to the bottom
- Inference: the mixture contains substances with different densities
- Technique used: centrifugation



Purifying materials

- Separating mixtures requires us to identify key physical properties then exploit the difference in properties to separate the components.
- In reality, the first step is to identify what are the components of the mixture. You will often be told.
- Determine the key physical properties of each component
- Pick a separation technique that exploits a difference in those properties



Separation by particle size

- Key property difference: Particle size
- Techniques: Sieving and filtration

sieving



Separating solids from solids

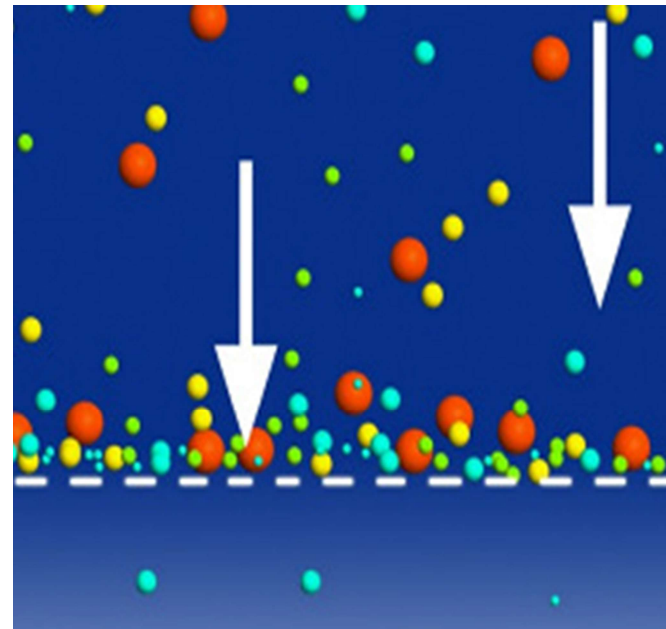
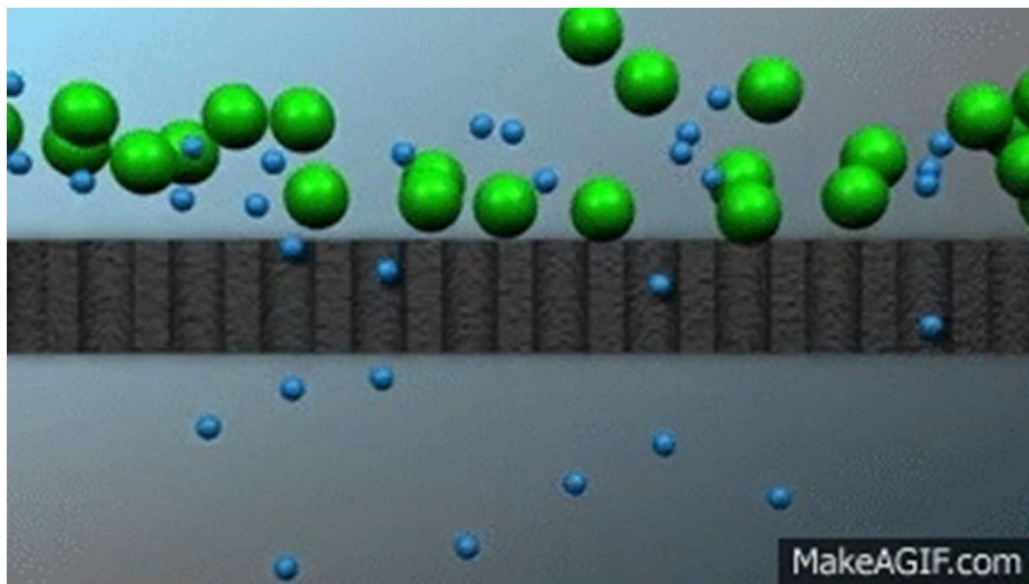
filtration



Separating solids from liquid or gases



Separation by particle size



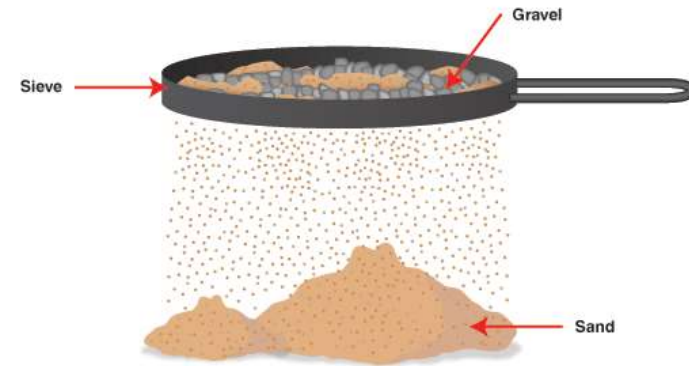
These particles are different sizes – only the small blue particles can pass through the barrier



Separation by particle size

Sieving – The process by which solid particles of various sizes can be physically separated using mesh of varying sizes.

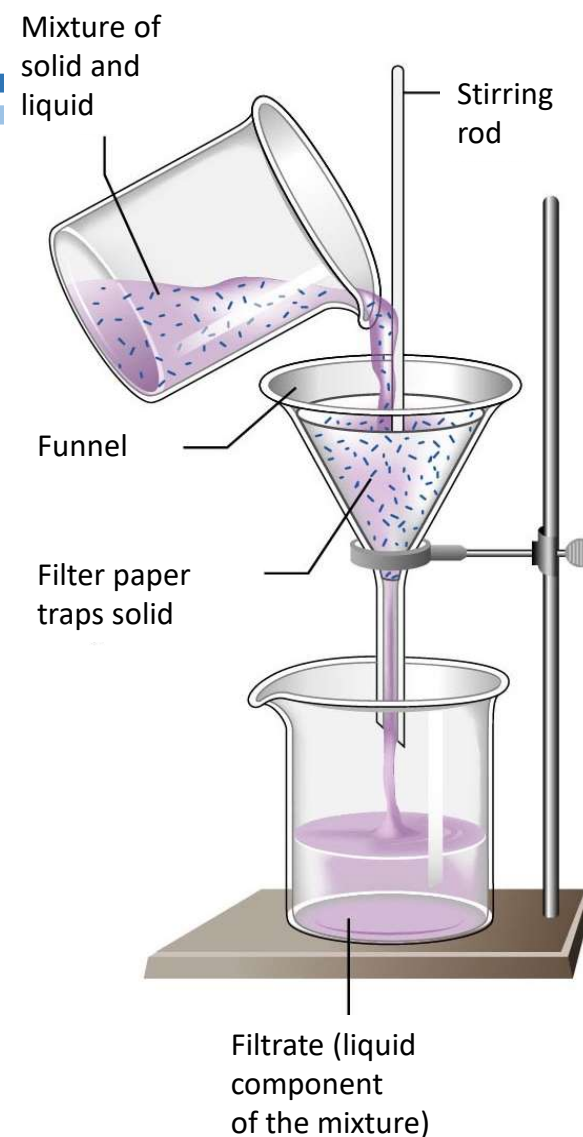
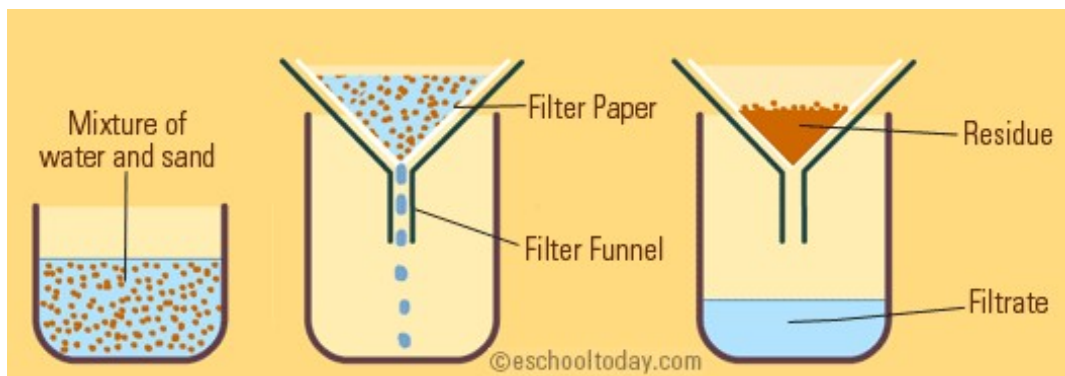
- sieving flour to remove any lumps or foreign material.
- Earth to separate sand and rocks





Separation by particle size

Filtration – Separation of solids or liquids from a liquid or a gas by using a barrier with holes smaller than the particles being separated.

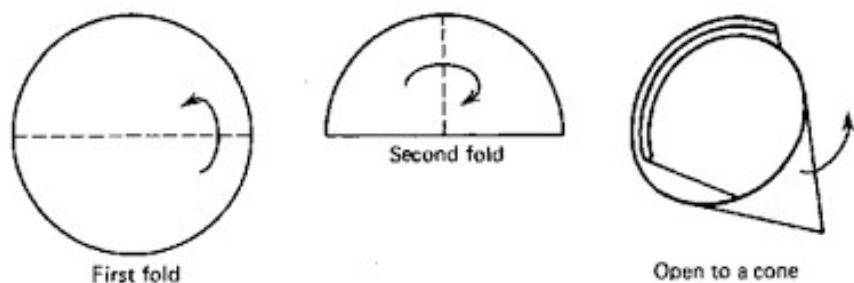




Separation by particle size

Folding filter paper – for lab-based filtrations

Cone-shaped filter

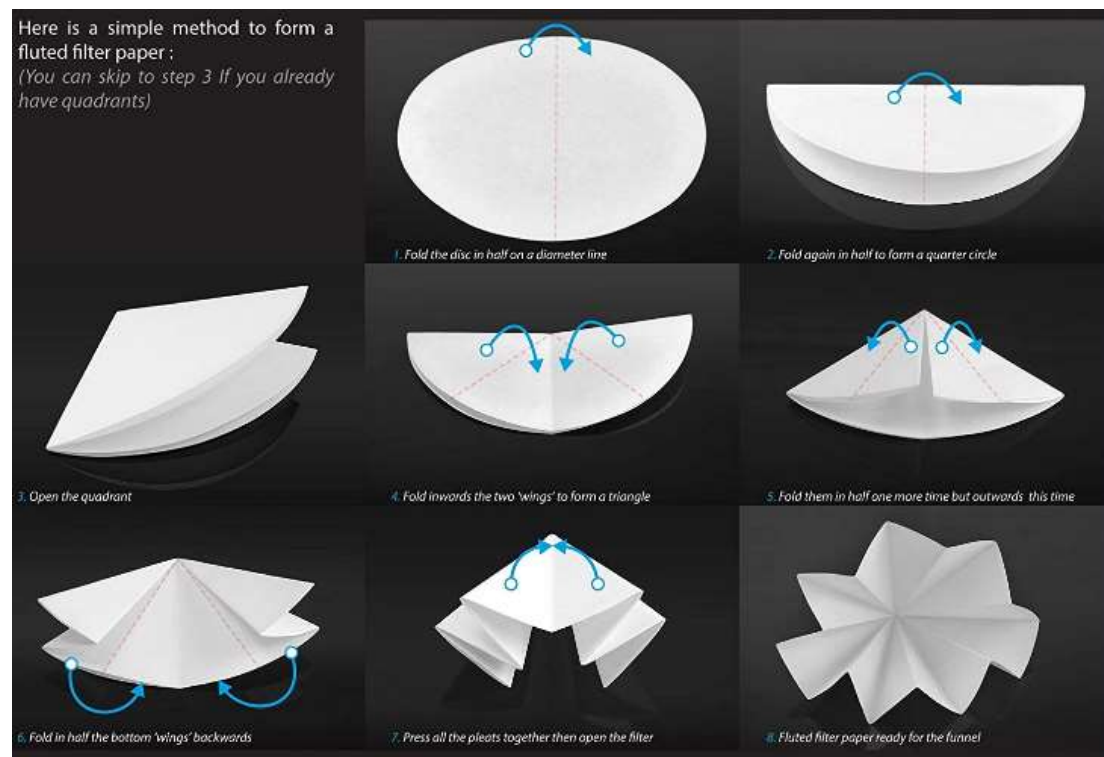


This video shows some of the ways to fold a filter paper (a skill you need for next week's experiment)

<https://www.youtube.com/watch?app=desktop&v=ctadASlwyT4>

Fluted filter

Here is a simple method to form a fluted filter paper:
(You can skip to step 3 if you already have quadrants)

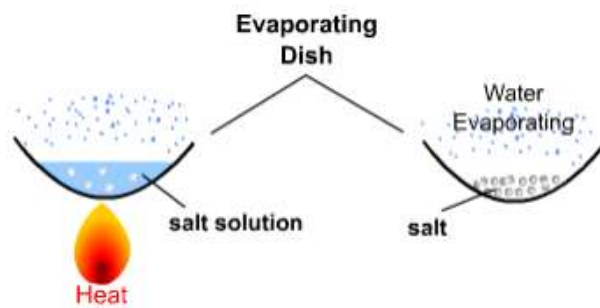




Separation based on boiling point

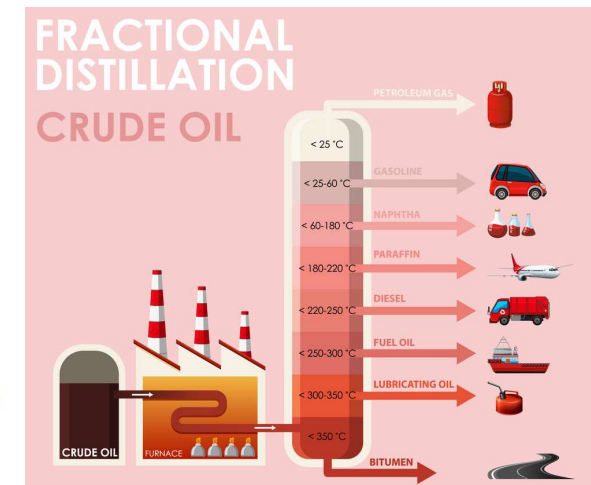
- Key property difference: Boiling point
- Techniques: evaporation and distillation

evaporation



Only want the high boiling substance

distillation

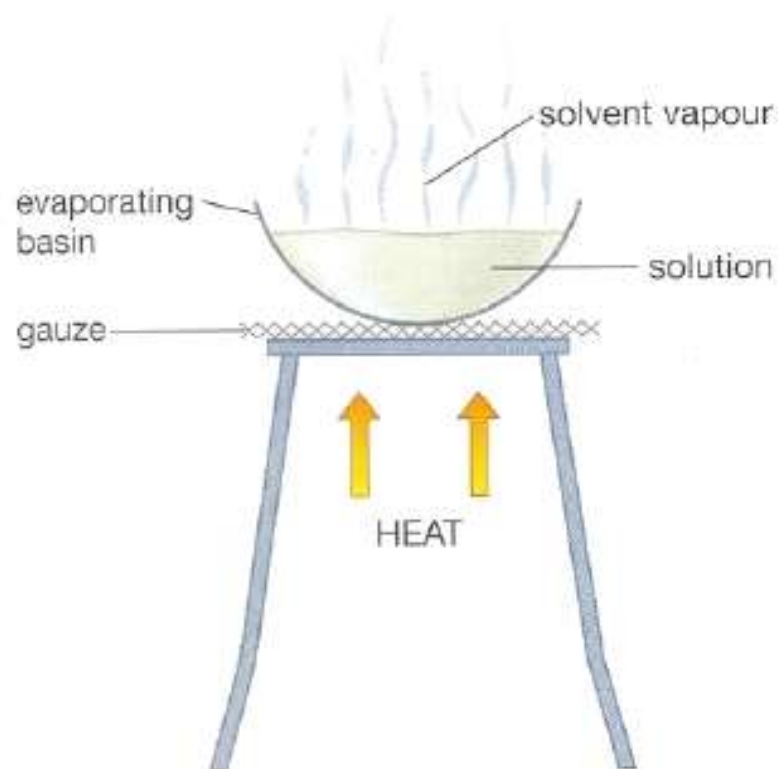
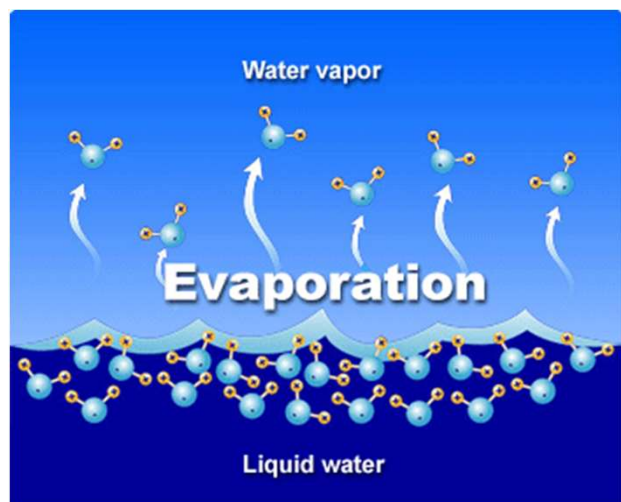


Want the low boiling substance or both



Separation based on boiling point

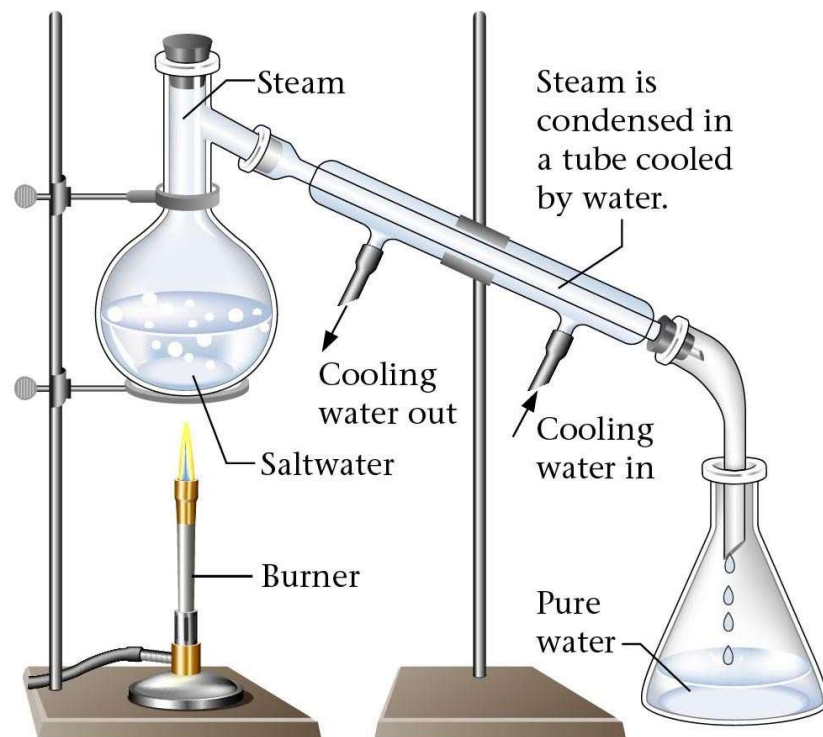
Evaporation – A process that uses heat to make a liquid solvent change state to a gas and leave behind the solute it had dissolved.





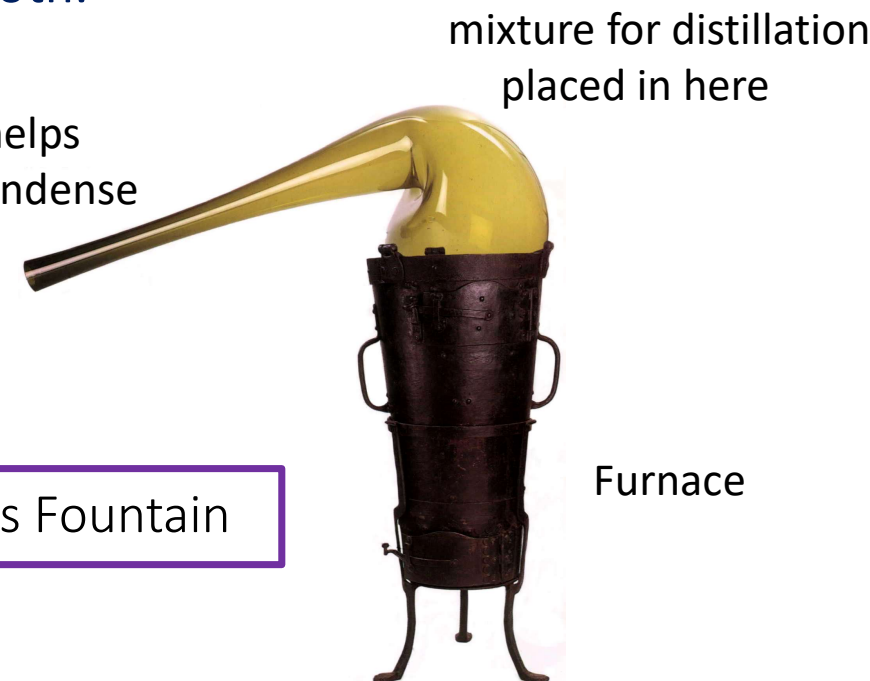
Separation based on boiling point

Distillation – A process that uses evaporation and condensation to separate solids from liquids or liquids from liquids, enabling the recovery of both.



long spout helps
vapors to condense

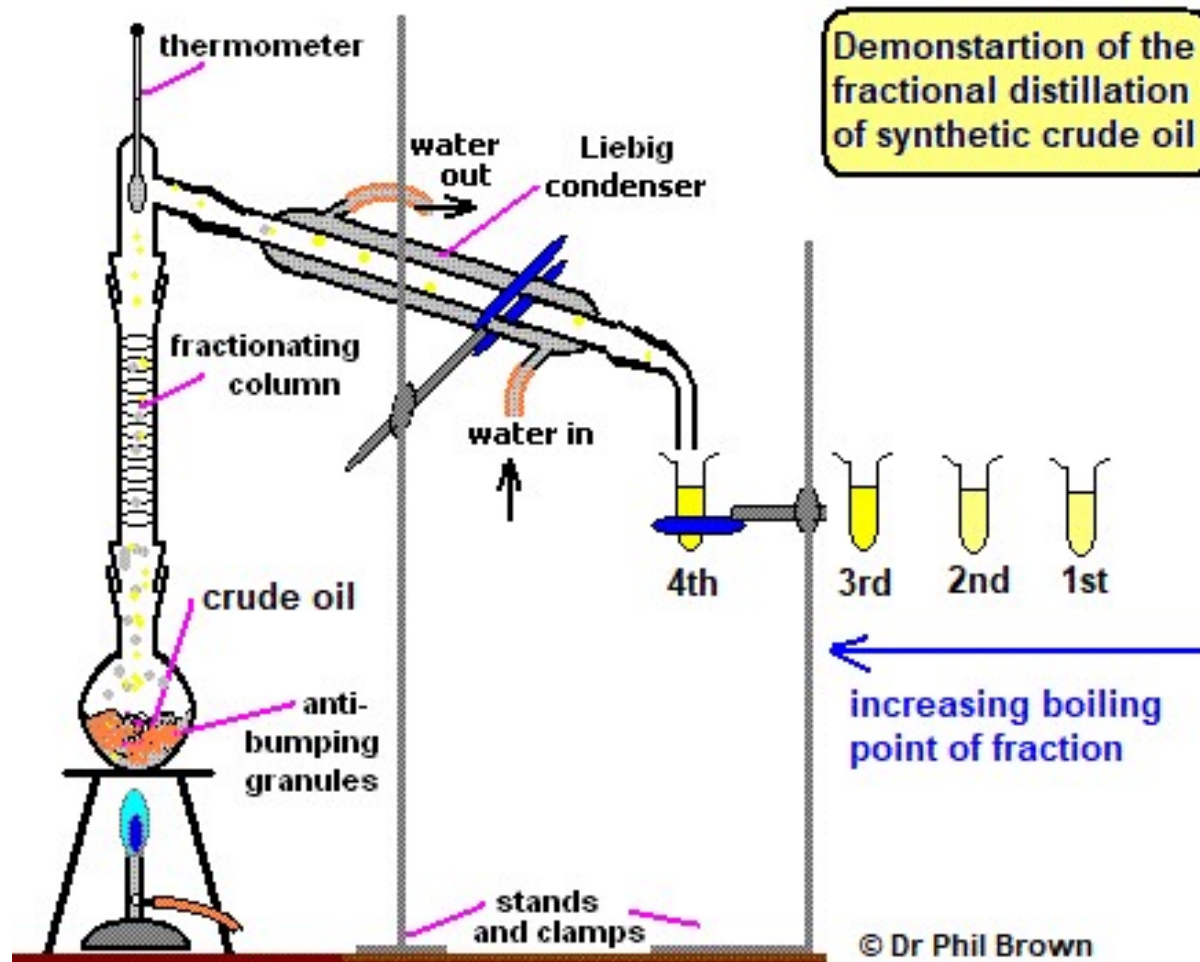
A Hero's Fountain





Separation based on boiling point

Fractional Distillation – use a long fractionating column to allow you to separate multiple fluids with different boiling points





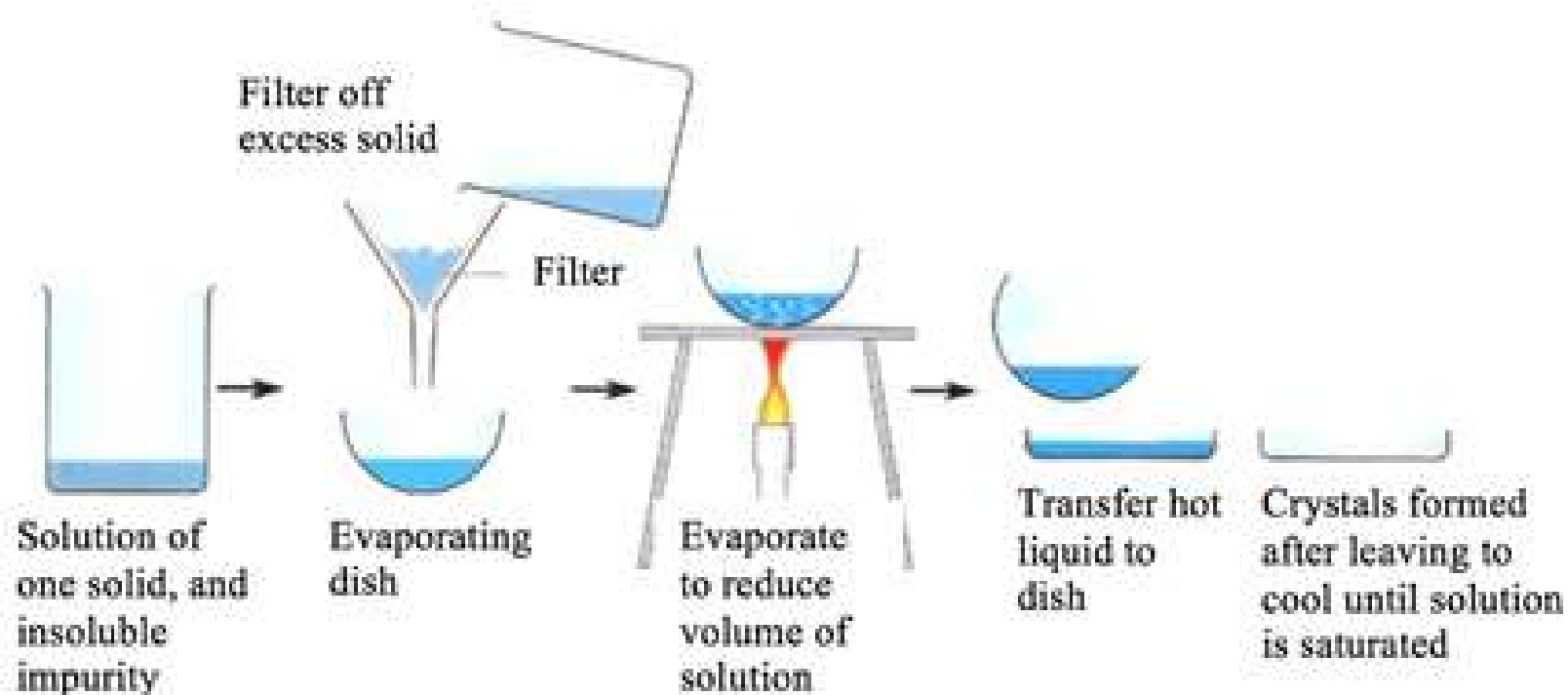
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Separation based on solubility

Crystallisation – Formation of crystals as a dissolved substance solidifies as the solvent is slowly evaporated.





Separation based on density

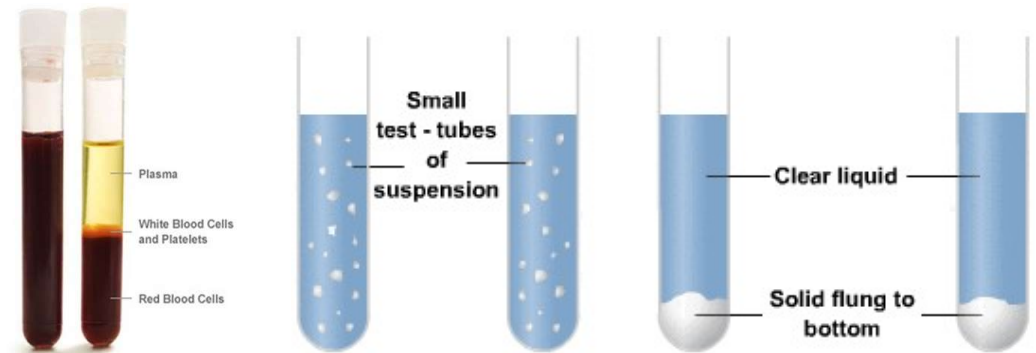
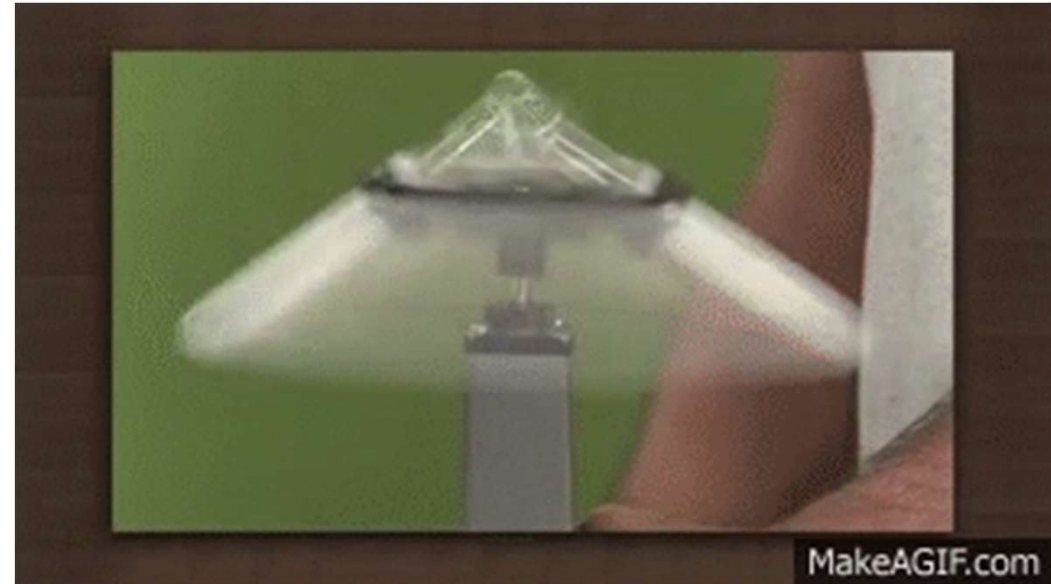
- Key property difference: Density
- Techniques: decantation, centrifugation, panning (all types of gravity separation)





Separation based on density

Centrifugation – Separation of solids from liquids, or separates liquids that have different weights, by using a device called a centrifuge. Spins very fast generating centrifugal force — the force from spinning that moves things away from the centre — For example, a centrifuge is used to separate blood cells from plasma cells.





Separation based on magnetism

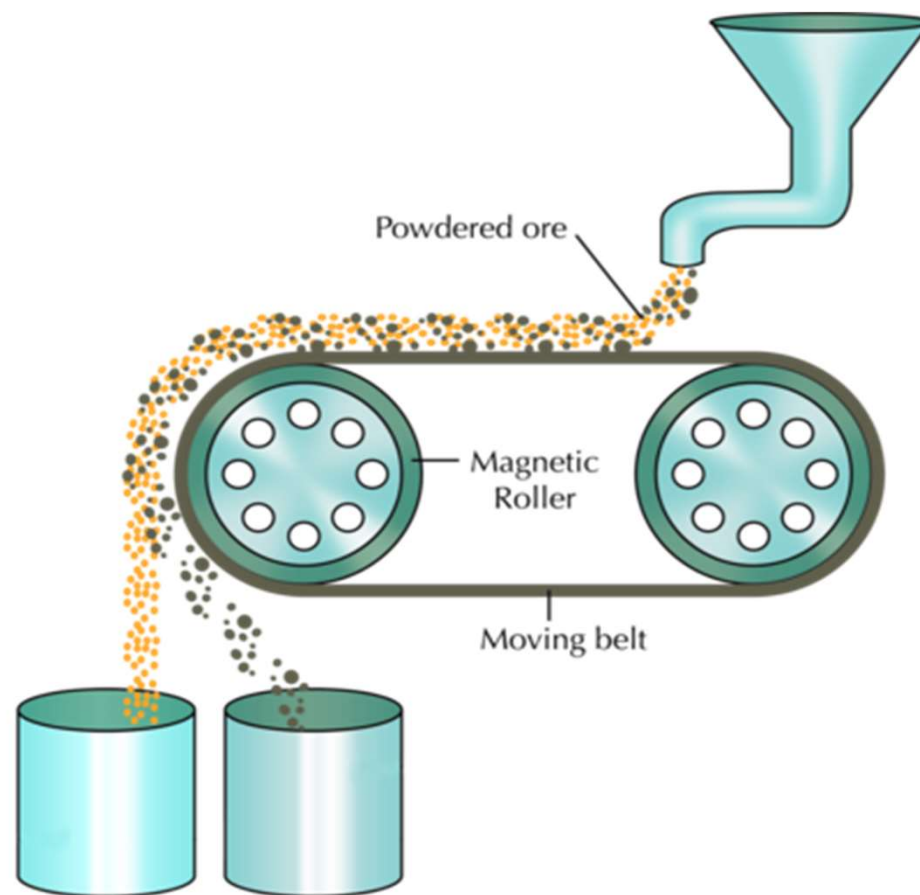
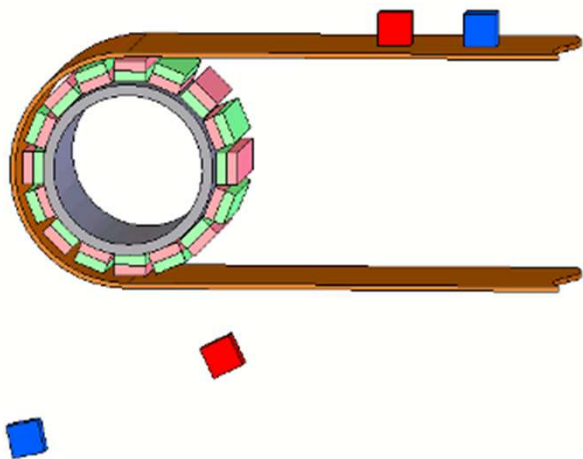
- Key property difference: Magnetic moment
- Techniques: magnetic separation





Separation based on magnetism

Magnetic separation – Magnetic separation is a process in which magnetically susceptible (e.g. some metals) material is extracted from a mixture using a magnetic force.





Separation flowcharts

Manual

You manually pick out the different materials



Chromatographic

Do this in detail in unit 2.

